



Opaque financial reports, R^2 , and crash risk[☆]

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ABSTRACT

We investigate the relation between the transparency of financial statements and the distribution of stock returns. Using earnings management as a measure of opacity, we find that opacity is associated with higher R^2 s, indicating less revelation of firm-specific information. Moreover, opaque firms are more prone to stock price crashes, consistent with the prediction of the Jin and Myers [2006. R^2 around the world: new theory and new tests. *Journal of Financial Economics* 79, 257–292] model. However, these relations seem to have dissipated since the passage of the Sarbanes–Oxley Act, suggesting that earnings management has decreased or that firms can hide less information in the new regulatory environment.

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1. Introduction

Financial economists and accountants have long viewed stock price changes as tied to new information about firms' prospects. However, Roll (1988) finds that only a relatively small portion of price movements can be explained by contemporaneous public news and speculates that traders acting on nonpublic firm-specific information could drive returns. These results have stimulated considerable interest in the relation between information and stock price dynamics and, in particular, between the R^2 from a modified index-model regression and the revelation of firm-specific news. For example, Morck, Yeung, and Yu (2000) show that R^2 is lower

(equivalently, firm-specific return variation is higher) in countries with greater property rights accorded to arbitrageurs who trade on firm-specific information. Following up on these insights, several other papers explore the connection between R^2 and various measures of opacity. Their common point of departure is the notion that greater transparency and more complete revelation of firm-specific information should reduce R^2 .

Opacity could, however, affect more than the second moment of stock return distributions. For example, Jin and Myers (2006) develop a model with incomplete transparency that predicts occasional (individual firm) stock price crashes as accumulated negative firm-specific information suddenly becomes publicly available. In fact, Jin and Myers find that several cross-country indicators of opacity predict crash risk. But, in comparison to the large R^2 literature, the empirical relation between opacity and crashes has been largely ignored.

In this paper, we further consider the empirical link between opacity and the distribution of stock returns. Our approach differs from the existing literature in two respects. First, instead of employing marketwide

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measures of opacity in cross-country comparisons, we develop a measure of opacity for individual firms based on an indicator of earnings management, specifically, the prior three years' moving sum of the absolute value of discretionary accruals. Second, we consider the empirical link between opacity and the crash risk of individual firms, a relation that is predicted by models such as that of Jin and Myers.

We demonstrate that our measure of opacity reliably predicts both R^2 and crash risk. While crash risk is associated with earnings management, the incidence of large positive jumps is not. Therefore, opacity does not predict fat-tailed distributions per se but is associated specifically with crashes. Finally, we show that in the post-Sarbanes-Oxley Act (SOX) years, the relations between discretionary accruals and both R^2 and crash risk essentially disappear. This pattern is consistent with the interpretation that opacity associated with earnings management has declined sharply in the post-SOX years.

These findings are significant for several reasons. First, they corroborate earlier research establishing a link between proxies for opacity and R^2 , but using more direct measures of opacity of particular firms. Second, they shed light on the process by which information is revealed to the market place. While it is well known that stock prices are more prone to big downward moves than upward ones (French, Schwert, and Stambaugh, 1987; Campbell and Hentschel, 1992; Bekaert and Wu, 2000), our findings indicate that this asymmetry is not due entirely to the exogenous stochastic process generating information but likely results as well from the way that firms manage the flow of information to capital markets. In particular, our findings suggest that part of the asymmetry is due to managers who are able to stockpile negative information, hiding it from investors' view until its accumulation reaches a tipping point sufficient to result in a stock price crash (see Kothari, Shu, and Wysocki, 2007). Third, an understanding of the firm-specific characteristics that can predict extreme outcomes could be useful in portfolio and risk-management applications that focus on tail events, for example, value at risk. Similarly, option pricing depends on skewness and crash risk. So-called smirk curves have characterized the implied volatility of individual stock options as well as index options since the crash of October 1987, and they are widely held to reflect risk of future crashes (see Dumas, Fleming, and Whaley, 1998 or Bates, 2000). Again, understanding the factors that drive cross-sectional variation in such tail risk would be of obvious importance to market participants and allow for sharper option pricing.

In the next section, we review the literature on opacity and stock returns and develop our hypotheses. Section 3 provides an overview of our data, and Section 4 presents tests of our hypotheses. Section 5 summarizes and concludes.

2. Opacity and stock return distributions

The link between opacity and R^2 has been extensively discussed in the literature. When less firm-specific

information is publicly available, fewer observable reasons exist for individual stock returns to depart from broad market indexes and stock market synchronicity increases. The link between opacity and crash risk has received comparatively much less attention.

Several mechanisms could engender crash risk or, more generally, negative skewness in returns. For example, it is well known that trading among investors who have different opinions could reveal the private signals of others and move prices even in the absence of new fundamental information (e.g., Romer, 1993). In Hong and Stein (2003), this process, combined with short sale constraints, imparts an asymmetry in which market declines differentially reveal the private signals of relatively pessimistic investors. Such revelation could lead other investors to downgrade their assessments of a firm's prospects, thereby reinforcing the decline.

Other sources of negative skewness focus on volatility feedback effects (e.g., French, Schwert, and Stambaugh, 1987 or Campbell and Hentschel, 1992). For example, big price movements could cause investors to reassess market volatility and increase required risk premia. An increased risk premium reduces equilibrium prices, which reinforces the impact of bad news but offsets the impact of good news, thus generating negative skewness.

Our empirical tests, however, are motivated by far simpler models of the information structure. They most directly follow the model of Jin and Myers (2006) and require neither disagreement among investors nor time variation in risk premia. Instead, the model envisions firm managers controlling at least a portion of the public access to fundamental information about the firm. Managers have incentives to stockpile bad news, but in some circumstances those incentives collapse, leading to a sudden release of accumulated negative information and a stock price crash. The management of firm-specific information simultaneously increases both R^2 and crash risk.

In Jin and Myers (2006), lack of full transparency concerning firm performance enables managers to capture a portion of cash flow, in the process absorbing (and making nonvisible) part of the variation in firm-specific performance. This increases R^2 . Managers are willing to personally absorb losses due to temporary bad performance to protect their jobs. However, following a run of sufficiently bad news, they are unwilling or unable to absorb any more losses; in other words, they have an abandonment option. If they abandon their positions, all of the hitherto unobserved negative firm-specific shocks become public at once, resulting in a crash. This process can give rise to long left tails in the return distribution.

Whereas the precise nature of opacity in Jin and Myers is largely irrelevant, Kirschenheiter and Melumad (2002) focus on a particular source of imperfect information: earnings smoothing. In their model, higher reported earnings increase the inferred level of permanent earnings and, therefore, the value of the firm. This effect is greater when reported earnings are perceived as more precise, and therefore managers generally have an incentive to smooth earnings, under-reporting earnings in response to positive surprises and over-reporting following negative surprises. But when news is particularly bad, the manager

under-reports earnings to the greatest extent possible, partially to reduce the inferred precision of the bad news and partially to enable shifting of discretionary income to future periods. This gives rise to occasional big baths, or stock price crashes. However, while the Kirschenheiter and Melumad model is consistent with stock price crashes, it does not predict a reduction in R^2 . Jin and Myers note that for R^2 to decrease, managers (or interpreting their model more generally, employees) must absorb some risk from shareholders. In the Kirschenheiter and Melumad framework, the timing of information releases is affected, but not the amount of risk borne by shareholders over extended periods.

Jin and Myers' cross-country empirical results indicate that measures of opacity in a given national market are associated with both higher average R^2 and higher crash risk of the firms in that country. But virtually all other empirical work on opacity has focused entirely on R^2 . Li, Morck, Yang, and Yeung (2004) also focus on country-level data, finding that capital market openness and better legal systems are associated with lower R^2 . Piotroski and Roulstone (2004) find that the presence of informed market participants such as institutional investors is associated with lower R^2 . Durnev, Morck, Yeung, and Zarowin (2003) show that stock returns better predict future earnings changes when R^2 is lower (when there is greater implied transparency). Wie and Zhang (2006) relate the secular decline in average R^2 between 1976 and 2000 to changes in the level and volatility of firm-specific information, particularly return on equity. Ferreira and Laux (2007) show that firms with fewer anti-takeover provisions display higher trading activity, better information about future earnings in stock prices, and higher levels of idiosyncratic returns.

Admittedly, not all of the literature is consistent. Ashbaugh, Gassen, and LaFond (2005) find that the Durnev, Morck, Yeung, and Zarowin (2003) results do not generalize in an international setting. Some papers suggest that lower R^2 could reflect greater noninformation-related noise in returns instead of more firm-specific information (e.g., West, 1988). Barberis, Shleifer, and Wurgler (2005) find that the simple addition or deletion of a firm to the Standard & Poor's (S&P) 500 index can significantly change its R^2 .

In contrast to most prior work, we employ a direct measure of opacity based on measures of earnings (specifically, accruals) management. Considerable evidence indicates that accruals management obscures at least some information about firm fundamentals (see, e.g., Sloan, 1996) and is thus a direct, firm-specific measure of opacity. In addition, aggressive earnings management is likely to proxy for management's general proclivity to hide information from the capital market and thus captures less easily quantifiable or observable aspects of opacity. This is an advantage of our empirical framework compared with that of Jin and Myers. In their cross-country setting, opacity is measured using characteristics of the broad capital market in which the firm happens to be situated. In contrast, our observations are at the firm level and allow us to construct proxies for the level of opacity chosen by each particular firm.

We take this choice as given and do not model cross-sectional differences in the underlying motivation of firms to manage their earnings. However, this is a topic that has received considerable attention in the corporate governance literature. Dechow, Sloan and Sweeney (1996) examine firms subject to enforcement actions by the Securities and Exchange Commission (SEC) and show that compared with a matched control sample these firms have weaker corporate governance. For example, they are less likely to have audit committees, less likely to have outside blockholders, more likely to have insiders holding a majority of the board seats, and more likely to have a chief executive officer who is also the chairman of the board. These authors also attempt to understand why firms manipulate earnings and conclude that "important motivations for earnings manipulation are the desire to raise external financing at low cost and to avoid debt covenant restrictions".

More recent research confirms many of these findings. For instance, Klein (2002) shows a negative relation between audit committee and board independence and earnings management. Cornett, Marcus, and Tehranian (2008) find that large stock ownership by institutional investors also constrains earnings management. Conversely, they show that economic rewards to a high stock price (in the form of incentive-based compensation such as option grants) encourage earnings management. Other studies also find that earnings management responds to payoffs associated with a temporarily high stock price (Bergstresser, Desai, and Rauh, 2006; Bergstresser and Philippon, 2006; Teoh, Welch, and Wong, 1998a, b). More generally, firms seem to respond to incentives to manipulate reported earnings to retain healthy market valuations. Specifically, Skinner and Sloan (2002) and Kinney, Burgstahler, and Martin (2002) show dramatic stock price declines associated with even small negative earnings surprises; Matsumoto (2002) and Burgstahler and Eames (2006) provide evidence of earnings management to meet analysts' forecasts.¹

We hypothesize that when a firm's financial reports are more opaque, less firm-specific information is available to affect its stock returns. Thus, we expect such firms to have stock returns with less idiosyncratic volatility relative to marketwide variation.

¹ While corporate governance and incentive-based compensation could be related to the propensity to manage earnings, we do not see an obvious concern with endogeneity. No obvious reason exists for firms with naturally higher or lower R^2 due to their lines of business to be more or less inclined to manage earnings. Therefore, we are not overly concerned about endogeneity or reverse feedback effects. In fact, while it is a stretch, one might argue that firms with high R^2 might be, if anything, less prone to manage earnings, which would cut against our hypothesis. This argument might follow from Collins and Kothari (1989), who find that high-beta firms (which tend to be high R^2 firms) exhibit lower price responses to earnings announcements, possibly because those earnings are capitalized at higher discount rates. The lower price response would reduce the incentive to manage earnings. In any event, our results stand up even when we measure earnings management relative to other firms in the same industry, for which all firms ought to have similar stock market synchronicity. This robustness also argues against a problem with endogeneity.

H1. *Firms with opaque financial reports have stock returns that are more synchronous with the market.*

However, these firms also face a higher likelihood of extreme outcomes when the firm-specific information is finally revealed to the market. Specifically, we hypothesize that firms managing reported earnings shelter bad information up to a point, but that once a threshold is crossed, the information comes out in one fell swoop, at which point a price crash is observed.

H2. *Firms with opaque financial reports face greater crash risk.*

In contrast, managers do not face incentives to shelter good information. So, we do not expect opaque financial reporting to be related to the likelihood of positive jumps.

H3. *Firms with opaque financial reports face no greater likelihood of positive discrete jumps in stock prices than more transparent firms.*

Finally, if Sarbanes-Oxley has been effective in lowering the level of earnings management via accrual manipulation, then our measure of opacity ought to be less highly related to R^2 and crash risk in the post-SOX era.

3. Sample development, variable measurement, and research design

This section presents the empirical methodology, including sample selection, variable definitions, and research design.

3.1. The sample

We combine firms' weekly stock return data from the Center for Research in Security Prices (CRSP) with annual financial data from Compustat. Weekly stock returns are assigned to each firm's fiscal year so as to match the time period of its reported financial data. Our sample period begins with fiscal year 1991, the first fiscal year for which we are able to estimate three annual lags of discretionary accruals using the Statement of Cash Flow method. Our sample period ends with the last year for which we have complete CRSP and Compustat data, fiscal year 2005. We begin with all firm-years on Compustat between 1991 and 2005 (109,026 firm-years). We exclude low-priced stocks (average price for the year <\$2.50); firm fiscal years with fewer than 26 weeks of stock-return data (22,777 firm-years are excluded); financial services firms and utilities (28,051 firm-years); and firm-years with insufficient financial data to calculate both three lags of discretionary accruals (9,903 firm-years) as well as control variables (7,413 firm-years). We are left with a final sample of 40,882 firm-years. The sample includes 43 of the 49 Fama and French industry definitions. The sample firm-years are approximately evenly distributed across our sample period. See Table 1 for details.

3.2. Accrual manipulation and opacity of financial reports

Under an accrual accounting system, the primary measure of firm performance is Earnings or Net Income, where Earnings is an estimate of all the current and expected future net cash flows associated with economic transactions transpiring during a given reporting period. Underlying the Earnings estimate are accruals, such as accounts receivable and depreciation, which are used to allocate revenues to the period in which they are earned and to match expenses to the revenues they generate. In contrast to Earnings, cash flows do not represent the creation of value but rather the distribution of value and thus they are not the primary measure of firm performance. Accruals must be added to actual cash flows to obtain the primary estimate of firm performance, Earnings.

This implies that reported earnings are necessarily estimates of firm performance, the accuracy of which depends on the quality of the accruals used to estimate the expected future net cash flows associated with past economic transactions. Earnings and underlying accruals are management's estimates of expected future net cash flows, as management has primary responsibility for developing the estimate of earnings reported to shareholders. Auditors attest that management's estimate have been developed within the confines of generally accepted accounting principles (GAAP). However, there is latitude within GAAP, and in some cases (e.g., WorldCom, Enron, etc.), management teams have ignored even those boundaries.

Thus, while, over the life of a firm, accruals must sum to zero and earnings must equal net cash flow, during particular periods earnings could deviate substantially from net cash flow and total accruals could reach significant magnitudes. In some instances, disparities between earnings and cash flows are reasonable, for example, resulting from growth in credit sales. These unbiased accruals are generally followed by cash flow realizations that remove or reverse the initial accrual (e.g., account receivables are decreased when cash is received from customers). In other instances, disparities between earnings and net cash flow result from earnings manipulation, and the inaccurate accruals are eventually reversed out by oppositely signed accruals instead of by cash flow realizations.

To distinguish between normal and discretionary accruals, we employ the modified Jones model (Dechow, Sloan, and Sweeney, 1995). Specifically, we estimate the following cross-sectional regression equation using the firms in each Fama and French industry for each fiscal year between 1988 and 2005:

$$\frac{TA_{jt}}{Assets_{jt-1}} = \alpha_0 \frac{1}{Assets_{jt-1}} + \beta_1 \frac{\Delta Sales_{jt}}{Assets_{jt-1}} + \beta_2 \frac{PPE_{jt}}{Assets_{jt-1}} + \varepsilon_{jt}, \quad (1)$$

where TA_{jt} denotes total accruals for firm j during year t , $Assets_{jt}$ denotes total assets for firm j at the end of year t ,

Table 1

Sample development, industry membership, and fiscal years of sample.

Source of data in Panel B is http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_49_ind_port.html.

Panel A: Sample development		Number of firm years	
All Compustat firm fiscal years 1991–2005		109,026	
Excluding firm fiscal years:			
With incomplete stock return data		22,777	
Financial services and utilities		28,051	
With insufficient financial data to calculate control variables		7,413	
With insufficient financial data to calculate three lags of discretionary accruals		9,903	
Final sample		40,882	
Panel B: Fama and French industries			
Industry	Number of firm years	Industry	Number of firm years
Aircraft	242	Consumer Goods	922
Agriculture	181	Measuring & Control Equipment	1,188
Automobiles & Trucks	882	Machinery	1,870
Beer & Liquor	181	Restaurants, Hotels, Motels	910
Construction Materials	1,066	Medical Equipment	1,610
Printing & Publishing	517	Non-Metallic & Industrial Metal Mining	223
Shipping Containers	173	Petroleum & Natural Gas	1,981
Business Services	2,260	Business Supplies	867
Chemicals	1,003	Personal Services	482
Electronic Equipment	3,019	Retail	2,682
Apparel	792	Rubber & Plastic Products	509
Construction	514	Shipbuilding, Railroad Equipment	107
Coal	37	Tobacco Products	53
Pharmaceutical Products	2,370	Candy & Soda	110
Electrical Equipment	846	Computer Software	2,851
Fabricated Products	247	Steel Works, etc.	912
Food Products	936	Communication	1,246
Entertainment	681	Recreation	433
Precious Metals	253	Transportation	1,295
Defense	80	Textiles	320
Computers	1,390	Wholesale	1,889
Healthcare	752		
Total			40,882
Panel C: Observations in each fiscal year			
Fiscal year	Number of observations		
1991	2,022		
1992	2,295		
1993	2,441		
1994	2,557		
1995	2,680		
1996	2,872		
1997	3,023		
1998	2,976		
1999	2,858		
2000	3,053		
2001	2,754		
2002	2,726		
2003	2,842		
2004	3,105		
2005	2,678		
Total	40,882		

$\Delta Sales_{jt}$ denotes change in sales for firm j in year t , and PPE_{jt} denotes property, plant, and equipment for firm j at the end of year t .²

Discretionary annual accruals as a fraction of lagged assets for firm j during year t ($DiscAcc_{jt}$) are then calculated using the parameter estimates from

² Total annual accruals equal income before extraordinary items minus cash flow from operating activities adjusted for extraordinary items and discontinued operations, specifically, TA_{jt} = Compustat annual

(footnote continued)
data items #123–[Compustat annual data item #308–Compustat annual data item #124].

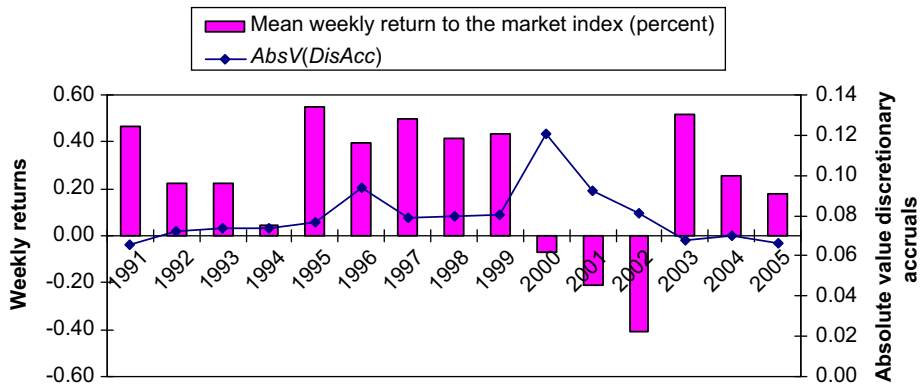


Fig. 1. Mean absolute value of discretionary accruals and mean weekly return on the Center for Research in Security Prices value-weighted market index.

Eq. (1):

$$DiscAcc_{jt} = \frac{TA_{jt}}{Assets_{jt-1}} - \left(\hat{\alpha}_0 \frac{1}{Assets_{jt-1}} + \hat{\beta}_1 \frac{\Delta Sales_{jt} - \Delta Receivables_{jt}}{Assets_{jt-1}} + \hat{\beta}_2 \frac{PPE_{jt}}{Assets_{jt-1}} \right), \quad (2)$$

where hats over the coefficients denote estimated values from regression Eq. (1). The inclusion of $\Delta Receivables_{jt}$ in Eq. (2) is the standard modification of the Jones (1991) model. This variable attempts to capture the extent to which a change in sales is due to aggressive recognition of questionable sales.

Dechow, Sloan, and Sweeney (1996) show the pattern of discretionary accruals for known manipulators, specifically, firms subject to enforcement actions by the SEC (see their Fig. 1 on p. 18). Discretionary accruals gradually increase as the alleged year of earning manipulation approaches and then exhibit a sharp decline. The increase in discretionary accruals is consistent with earnings manipulation; the decline, with the reversal of prior accrual overstatements. The large, positive discretionary accruals recorded by these firms are followed by large, negative discretionary accruals instead of by the positive cash flow realizations that would have tended to follow unbiased accrual practice. Dechow, Sloan, and Sweeney also show that these firms generally manipulate reported earnings from one to three years before being detected (see their Table 3) and that the overstated accruals of these firms typically reverse fairly quickly, with negative discretionary accruals following the prior positive ones in the years immediately following the periods of earnings manipulation.

Taken together, these facts motivate a simple, but intuitively appealing, measure of opacity in financial reports, the three-year moving sum of the absolute value of annual discretionary accruals:

$$OPAQUE = AbsV(DiscAcc_{t-1}) + AbsV(DiscAcc_{t-2}) + AbsV(DiscAcc_{t-3}). \quad (3)$$

The straightforward idea behind this measure is that firms with consistently large absolute values of discretionary

accruals are more likely to be managing reported earnings, thus revealing less firm-specific information to investors. For example, the characteristic pattern of manipulation shown in Dechow, Sloan, and Sweeney—large positive followed by large negative abnormal accruals—would result in a high measure of earnings management, as both the positive and negative abnormal values would contribute to the moving sum of absolute discretionary accruals. We use a three-year moving sum (instead of a one-year value) to capture the multi-year effects of earnings management and because the moving sum is more likely to reflect an underlying policy of the firm to manage earnings.³

Figs. 1 and 2 plot the cross-sectional means of $AbsV(DiscAcc)$ and $OPAQUE$, respectively, over the sample period. We also include the mean weekly return on the CRSP value-weighted market index in each year to examine the relation between levels of earnings management and market returns.

Fig. 1 shows that the mean of $AbsV(DiscAcc)$ is usually between 6% and 8% of lagged total assets. However, in four years (1996, 2000, 2001, and 2002), the mean level of $AbsV(DiscAcc)$ exceeds this typical range, and in three out of four of these high-manipulation years the mean weekly return on the market index is negative (aggregating to annual returns of −3.6%, −10.5% and −19.3% in 2000, 2001, and 2002, respectively). Thus, Fig. 1 suggests that the level of earnings management could be related to overall market performance. Fig. 2 demonstrates that our primary measure of earnings management, $OPAQUE$, is less highly correlated with overall market performance. Nevertheless, we control for the level of the overall market performance in our analyses below.⁴

³ We repeat our analysis substituting latest-year absolute discretionary accruals for the moving sum (see Table 6, Panel B). While results were qualitatively similar, the moving sum was a more powerful predictor of R^2 as well as crash propensity.

⁴ In addition, to ensure that our findings are not sensitive to prevailing market conditions, we re-run all regression analyses using a Fama and MacBeth approach with Pontiff (1996) adjusted standard errors. The findings are qualitatively unchanged.

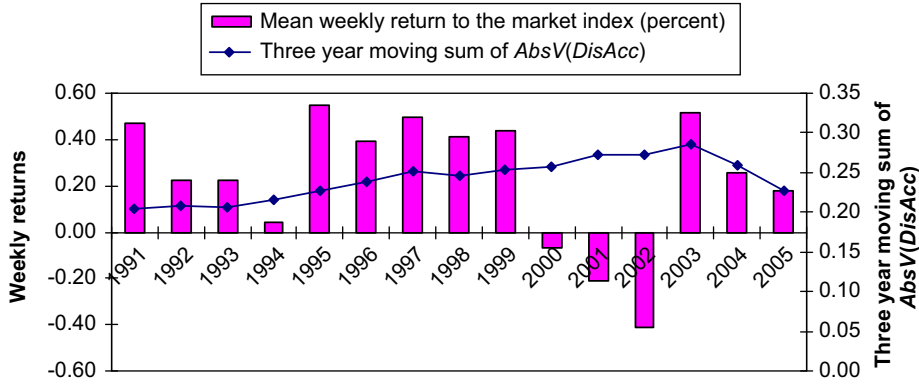


Fig. 2. OPAQUE and mean weekly return on the Center for Research in Security Prices value-weighted market index.

3.3. Measuring idiosyncratic risk

R^2 s and residual returns are calculated from an expanded index model regression:

$$r_{j,t} = \alpha_j + \beta_{1j}r_{m,t-1} + \beta_{2j}r_{i,t-1} + \beta_{3j}r_{m,t} + \beta_{4j}r_{i,t} + \beta_{5j}r_{m,t+1} + \beta_{6j}r_{i,t+1} + \varepsilon_{j,t}, \quad (4)$$

where $r_{j,t}$ is the return on stock j in week t , $r_{m,t}$ is the CRSP value-weighted market index, and $r_{i,t}$ is the Fama and French value-weighted industry index. We allow for nonsynchronous trading by including lead and lag terms for the market and industry indexes (Dimson, 1979).⁵

The residuals from Eq. (4) are highly skewed. We transform them to a roughly symmetric distribution by defining the *Firm-Specific Weekly Return* as the log of one plus the residual return from Eq. (4). This transformation allows us to define crash and positive jumps symmetrically, as residual returns corresponding to a threshold number of standard deviations either above or below the mean.⁶

We define crash and jump events using firm-specific returns, i.e., using residuals from Eq. (4). Using actual returns would result in an abundance of crashes during broad market declines and jumps during advances. The firm-specific crashes and jumps of interest to us require that we define extreme events using residual returns.

$1-R^2$ is a natural measure of firm-specific volatility or (lack of) market synchronicity, but it is bounded between zero and one, which creates complications for empirical estimation. We follow common practice (e.g., Morck, Yeung, and Yu, 2000) in defining idiosyncratic risk using a logistic transformation of R^2 , which can range from negative to positive infinity.

$$IDIOSYN = \ln\left(\frac{1-R^2}{R^2}\right). \quad (5)$$

⁵ Our findings below are qualitatively unchanged if we include either zero leads and lags or two leads and two lags in the expanded index model. Finally, using bi-weekly returns instead of weekly returns also leaves our findings qualitatively unchanged.

⁶ This transformation does seem to make the distribution essentially symmetric, even in the tails, which is our greatest concern. For example, the 99th percentile of the distribution is 0.186 while the first percentile is -0.184 .

Thus, a high value for *IDIOSYN* indicates a high level of idiosyncratic risk.

As discussed in Section 2, we expect firms with more opaque financial reports to have stock returns that are more synchronous with the market. Opacity implies that less firm-specific information is available to affect the firm's stock returns, and thus the firm exhibits lower values of *IDIOSYN*. However, while opaque firms could generally tend to follow the market more closely, they also present a higher likelihood of an extreme outcome or crash when accumulated negative firm-specific information is finally revealed to investors.

3.4. Measuring the frequency of crashes and jumps

We construct measures of the likelihood of crashes or positive jumps based on the number of the *Firm-Specific Weekly Returns* exceeding 3.09 standard deviations below or above its mean value, respectively, with 3.09 chosen to generate a frequency of 0.1% in the normal distribution.⁷ An indicator variable, *CRASH*, is set equal to one for a firm-year if the firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 standard deviations below the mean weekly firm-specific return for that fiscal year; otherwise, *CRASH* is set equal to zero. Similarly, *JUMP* is an indicator variable defined as one if a firm experiences one or more *Firm-Specific Weekly Return* 3.09 standard deviations above the mean value for that fiscal year and zero otherwise.

This definition of a crash implies a substantial weekly return. For example, the average standard deviation of $\ln(1+\text{residual return})$ across our sample is 5.8%. Using this value, a negative 3.09-sigma event translates into an abnormal weekly return of -18% .

Given our definition of a crash, if firm-specific returns were normally distributed, one would expect to observe 0.1% of the sample firms crashing in any week. A similar number of jumps would be expected. The probability of a

⁷ While the mean residual from the index model regression, Eq. (4), is zero by construction, the mean value of $\ln(1+\text{residual return})$ is not. Therefore, we calculate deviations from the firm-year mean of $\ln(1+\text{residual return})$ to define crashes and positive jumps.

Table 2

Crashes and jumps, frequency and weekly returns.

Panel A reports the frequency of firm-specific crashes and jumps for 40,882 firm-years in the sample period 1991–2005. Crashes and jumps are defined based on residuals from an expanded index model regression with market and industry returns as explanatory variables. Weekly firm-specific residual returns that are 3.09 standard deviations below or above the mean for the firm's fiscal year are categorized as crashes or jumps.

Panel B reports the mean, median, and variance of raw weekly returns for *CRASH* weeks, *JUMP* weeks, and *ALL OTHER* weeks. The first set of columns presents statistics for individual firms. The middle set of columns presents analogous results for the market index. In this panel, *CRASH* (or *JUMP*) weeks refer to any week in which any firm in the sample crashes (or jumps). The last set of columns report statistics averaging across industries. If any firm in an industry crashes (or jumps) in a given week, that is defined as a crash (or a jump) week for the industry. Weekly returns for 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

Panel A: Crash and jump frequency										
Frequency		Number of observations					Percent of sample			
Crashes in the firm year										
0		33,932					83.0			
1		6,766					16.6			
2		184					0.5			
3		0					0.0			
Total		40,882								
Jumps in the firm year										
0		31,853					77.9			
1		8,767					21.4			
2		261					0.6			
3		1					0.0			
Total		40,882								
Panel B: Return in crash and jump weeks versus all other weeks										
Week classification	Number of observations	Firm returns			Market index			Industry index		
		Mean	Median	Variance	Mean	Median	Variance	Mean	Median	Variance
CRASH Weeks	7,134	−0.2274	−0.2014	0.0147	0.0033	0.0039	0.0004	0.0020	0.0020	0.0010
JUMP Weeks	9,292	0.3327	0.2623	0.1007	0.0055	0.0066	0.0003	0.0084	0.0071	0.0008
ALL OTHER Weeks	2,101,147	0.0026	0.0000	0.0054	0.0023	0.0036	0.0005	0.0025	0.0029	0.0010

crash or a jump over the course of a year would then be $1 - (1 - 0.001)^{52} = 0.0507$. In fact, we observe considerably greater frequency of both crashes and jumps than this benchmark.⁸ Table 2, Panel A, indicates that 17.1% of the firm-years in our sample experience at least one *CRASH* (6,950 firm-years); 22.0% experience at least one *JUMP* (9,029 firm-years).

In Table 2, Panel B, we highlight the mean, median, and variance of raw weekly returns for three subsamples of firm-weeks: *CRASH* weeks, *JUMP* weeks, and *ALL OTHER* weeks. The first set of columns presents statistics for individual firms. The mean weekly return for *CRASH* weeks is −22.74%, for *JUMP* weeks, 33.27%, and for *ALL OTHER* weeks, 0.26%. Median returns demonstrate a similar pattern. As expected, the variance of weekly returns is dramatically higher for *CRASH* and *JUMP* weeks than for *ALL OTHER* weeks. The middle set of columns presents analogous results for the market index. In this panel, *CRASH* (or *JUMP*) weeks refer to any week in which any firm in the sample crashes (or jumps). Even though *CRASH* and *JUMP* weeks are categorized using residuals from an expanded index model, the mean and median

weekly return to the market index is generally a bit higher in *JUMP* weeks than in *ALL OTHER* weeks in the sample. This is not the case for *CRASH* weeks. Nevertheless, the variance of the market index is not higher in *CRASH* and *JUMP* weeks. Finally, the last set of columns report statistics averaging across industries. If any firm in an industry crashes (or jumps) in a given week, that is defined as a crash (or a jump) week for the industry. Industry results indicate slightly lower (higher) mean and median returns in *CRASH* (*JUMP*) weeks, but equivalent variance.

Figs. 3 and 4 highlight the frequency of *CRASH*es and *JUMP*s, respectively, across the sample years. We include the value-weighted market index in Figs. 3 and 4 to demonstrate that the frequency of these extreme outcomes is in fact independent of market movements, which is not surprising given that we employ residuals from an index model to define *CRASH*es and *JUMP*s. The frequency of *CRASH*es is significantly higher in the second half of the sample period than in the first half, whereas the frequency of *JUMP*s is largely unchanged over the time period.

3.5. Control variables

Following prior literature, our control variables include *SIZE*, defined as the log of the market value of equity at the

⁸ Given our research question, we do not expect the *Firm-Specific Weekly Returns* to be normally distributed. We merely use the 0.1% cutoff of the normal distribution as a convenient way to obtain reasonable benchmarks for extreme events, *CRASH*es and *JUMP*s.

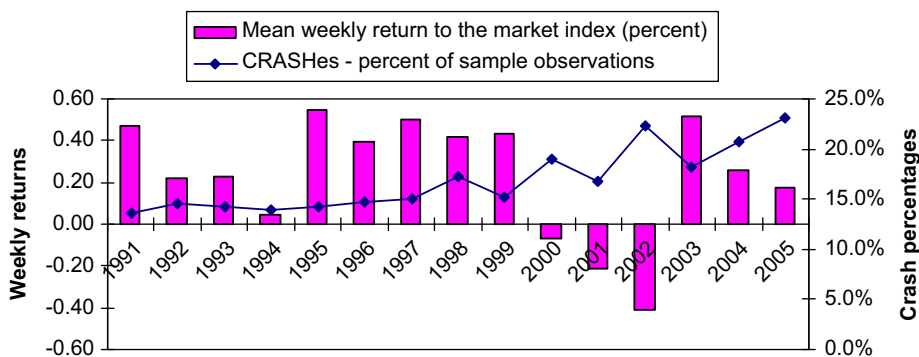


Fig. 3. Crash frequency and the mean weekly return on the Center for Research in Security Prices value-weighted market index.

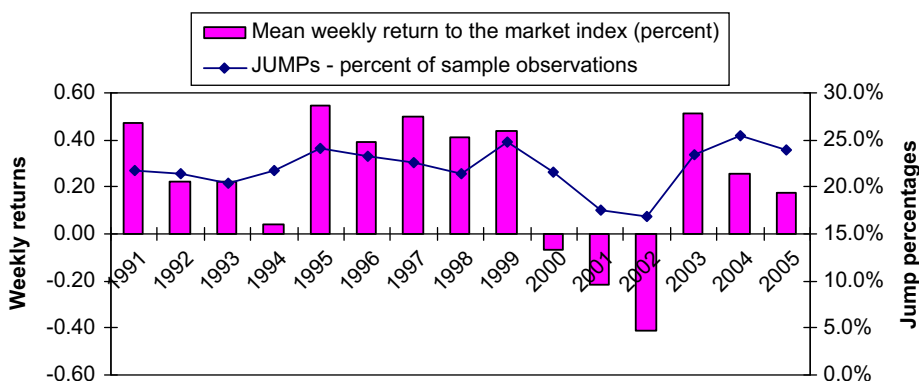


Fig. 4. Jump frequency and the mean weekly return on the Center for Research in Security Prices value-weighted market index.

beginning of the fiscal year; M to B , defined as the ratio of the market value of equity to the book value of equity measured at the beginning of the fiscal year; LEV , defined as the book value of all liabilities scaled by total assets, again measured at the beginning of the fiscal year; and finally, contemporaneous ROE , defined as income before extraordinary items divided by the book value of equity.

We include the variance of the Fama and French weekly industry index (VAR_INDRET) as an additional control variable in regressions with $IDIOSYN$ as the dependent variable because higher industry variance increases systematic risk, and hence the R^2 of Eq. (4). Finally, following Jin and Myers (2006), we also include the $KURTOSIS$ and the $SKEWNESS$ of the Firm-Specific Weekly Returns as control variables in some of our $IDIOSYN$ regressions.

3.6. Descriptive statistics and correlation matrix

Table 3, Panel A, presents descriptive statistics for our key variables of interest; Table 3, Panel B, presents the correlation matrix. Pearson correlations appear above the diagonal, and Spearman correlations below. $OPAQUE$ is negatively correlated with R^2 , and positively correlated with tail events, as measured by skewness, kurtosis, crash

probability, or jump probability.⁹ Not surprisingly, the various measures of fat tails, i.e., kurtosis, crash risk, and jump probability, are all positively correlated. Size is negatively correlated with opacity and jump probability and positively correlated with R^2 and crash probability. As expected, R^2 is negatively correlated with kurtosis, skewness, crash risk, and jump probability in this univariate analysis.

3.7. Opacity and information hoarding

Before turning to our main results, which connect our measure of opacity to R^2 and crash risk, we pause to first corroborate our interpretation of our opacity measure. We wish to show that firms with high levels of $OPAQUE$ are less transparent and, in particular, are more likely to

⁹ The negative correlation between R^2 and $OPAQUE$ appears to contradict one of our main hypotheses. However, after controlling for other characteristics, in particular size, the correlation is strongly positive. For example, we show in Table 5 that, once firms are stratified by size, R^2 increases with $OPAQUE$ for each size group. The multiple regressions in Table 6 also confirm the positive relation between R^2 and opacity, after controlling for firm size. The univariate correlation (that does not control for size) appears to be negative because large firms tend to have higher R^2 s and to be less opaque, as shown in Table 5, Panels B and C.

Table 3

Key variables of interest.

Panel A covers 40,882 firm-years in the sample period 1991–2005. Q1 and Q3 are first and third quartile values. In Panel B, correlations are computed from 40,882 firm-years in the sample period 1991–2005. Pearson correlations are above the diagonal; Spearman correlations below the diagonal. *p*-values appear below correlations. See Appendix A for variable definitions.

Panel A: Descriptive statistics												
Variable		Q1		Mean		Median		Q3		Standard deviation		
Market value (lagged)		63.2		3,157.8		258.0		1,191.5		14,300.5		
Opaque		0.102		0.243		0.174		0.302		0.222		
R ²		0.128		0.250		0.209		0.331		0.164		
Std Dev [ln(1+residual)]		0.036		0.058		0.052		0.074		0.030		
Kurtosis		0.152		1.672		0.878		2.146		2.730		
Skewness		−0.258		0.113		0.134		0.523		0.776		
ROE		−0.006		−0.011		0.087		0.155		0.426		
Market-to-book (lagged)		1.240		3.077		2.028		3.453		3.434		
Leverage (lagged)		0.308		0.475		0.484		0.634		0.213		
Var(Industry Index)		0.0004		0.0010		0.0007		0.0012		0.0010		
Panel B: Correlation matrix												
Variable	Size	Opaque	R ²	Std Dev ln(1+resid)	Kurtosis	Skewness	Crashes	Jumps	ROE	Market to book	Leverage	Variance of Industry Index
Size (lagged)		−0.215 <0.0001	0.580 <0.0001	−0.465 <0.0001	−0.083 <0.0001	−0.210 <0.0001	0.076 <0.0001	−0.171 <0.0001	0.182 <0.0001	0.245 <0.0001	0.142 <0.0001	0.048 <0.0001
Opaque	−0.282 <0.0001		−0.074 <0.0001	0.358 <0.0001	0.035 <0.0001	0.030 <0.0001	0.004 0.4048	0.034 <0.0001	−0.216 <0.0001	0.182 <0.0001	−0.113 <0.0001	0.129 <0.0001
R ²	0.548 <0.0001	−0.100 <0.0001		−0.305 <0.0001	−0.179 <0.0001	−0.117 <0.0001	−0.024 <0.0001	−0.157 <0.0001	0.096 <0.0001	0.151 <0.0001	0.072 <0.0001	0.176 <0.0001
Std Dev ln(1+resid)	−0.511 <0.0001	0.407 <0.0001	−0.311 <0.0001		0.236 <0.0001	0.038 <0.0001	0.082 <0.0001	0.131 <0.0001	−0.388 <0.0001	0.116 <0.0001	−0.129 <0.0001	0.309 <0.0001
Kurtosis	−0.129 <0.0001	0.070 <0.0001	−0.215 <0.0001	0.224 <0.0001		−0.098 <0.0001	0.430 <0.0001	0.373 <0.0001	−0.064 <0.0001	−0.008 0.1074	−0.027 <0.0001	0.002 0.6691
Skewness	−0.239 <0.0001	0.048 <0.0001	−0.163 <0.0001	0.077 <0.0001	0.086 <0.0001		−0.597 <0.0001	0.556 <0.0001	−0.001 0.8881	−0.073 <0.0001	0.014 0.0063	−0.032 <0.0001
Crashes	0.088 <0.0001	0.010 0.0398	−0.015 0.0023	0.082 <0.0001	0.486 <0.0001	−0.565 <0.0001		−0.149 <0.0001	−0.023 <0.0001	0.034 <0.0001	−0.021 <0.0001	0.021 <0.0001
Jumps	−0.175 <0.0001	0.042 <0.0001	−0.169 <0.0001	0.119 <0.0001	0.535 <0.0001	0.592 <0.0001	−0.149 <0.0001		−0.034 <0.0001	−0.052 <0.0001	−0.003 0.5510	−0.025 <0.0001
ROE	0.281 <0.0001	−0.157 <0.0001	0.148 <0.0001	−0.374 <0.0001	−0.100 <0.0001	−0.025 <0.0001	−0.030 <0.0001	−0.048 <0.0001		−0.062 <0.0001	0.034 <0.0001	−0.099 <0.0001
M/B ratio (lagged)	0.422 <0.0001	0.097 <0.0001	0.242 <0.0001	−0.030 <0.0001	−0.051 <0.0001	−0.131 <0.0001	0.068 <0.0001	−0.097 <0.0001	0.294 <0.0001		0.059 <0.0001	0.090 <0.0001
Leverage (lagged)	0.136 <0.0001	−0.142 <0.0001	0.061 <0.0001	−0.157 <0.0001	−0.031 <0.0001	0.008 0.0991	−0.022 <0.0001	−0.004 0.4767	0.127 <0.0001	−0.020 <0.0001		−0.090 <0.0001
Variance of Industry Index	0.045 <0.0001	0.170 <0.0001	0.177 <0.0001	0.309 <0.0001	0.012 0.0162	−0.036 <0.0001	0.022 <0.0001	−0.024 <0.0001	−0.092 <0.0001	0.027 <0.0001	−0.075 <0.0001	

Table 4

Opacity and financial restatements.

This table shows the relation between opacity and the incidence of restatements of financial statements. Panel A divides firm-years into three groups based on the value of *OPAQUE* and presents the rate of restatements for each group. Panel B compares the levels of *OPAQUE* for firm-years with and without restatements.

Panel A: Incidence of restatements by opacity measure			
Opacity group		Percent of observations with restatements	
Lowest		1.20%	
Middle		1.63%	
Highest		1.92%	
Chi ² statistic for equality across groups		8.35 (<i>p</i> = 0.0154)	
Chi ² statistic for equality between high and low opacity groups		8.37 (<i>p</i> = 0.0038)	
Panel B: Levels of opacity for restatement and no-restatement firm-years			
	Number of observations	Mean <i>OPAQUE</i>	Median <i>OPAQUE</i>
No restatements	14,437	0.2645	0.1862
Restatements	233	0.3634	0.2183
<i>t</i> -Statistic for equality across groups		2.60	3.75
<i>p</i> -value		0.0098	< 0.0001

withhold negative news from investors.¹⁰ To do so, we examine the relation between *OPAQUE* and the frequency of financial statement restatements. We obtain a sample of such restatements from Cheng and Farber (2008), who in turn developed their sample from 919 restatements of public companies between January 1, 1997 and June 30, 2002 identified in a US General Accounting Office (GAO) report. US General Accounting Office (2002) focuses only on restatements connected to “accounting irregularities”, i.e., “an instance in which a company restates its financial statements because they were not fairly presented in accordance with generally accepted accounting principles (GAAP). This would include material errors and fraud”. Thus, these are fairly extreme and relatively uncommon events that indicate less-than-forthright accounting practice.

We are able to match 233 of the 919 restatements with our subsample of 14,670 firm-years with fiscal years 1997–2001 and find that the incidence of financial restatements in this subsample is related to our opacity measure. We divide the subsample into three groups based on firm-year opacity level. Table 4, Panel A, shows that the rate of restatements increases with opacity. Only 1.2% of the firm-years in the low-opacity group experience a restatement compared with nearly 2% of the firm-years falling in the high-opacity group. This difference is statistically significant at a 0.0038 level. Moreover, differences in restatement rates become more extreme when we cut the data more finely. For example, if we split our sample into opacity deciles, the incidence of restatements in the highest decile is triple that in the lowest, 3% versus 1%.¹¹

A more dramatic view of the association between opacity and restatements comes from Panel B of Table 4, which compares mean and median values of *OPAQUE* in the group of firm-years with restatements to those in the no-restatement group. The restatement group demonstrates significantly higher levels of opacity. The difference in mean values of *OPAQUE* is 0.0989 (= 0.3634–0.2645). Because this is the mean value of a three-year moving sum, it corresponds to a difference of 0.033 in the mean annual absolute discretionary accrual. Discretionary accruals are expressed as a fraction of firm assets. Therefore, this value corresponds to a swing in measured annual return on assets of fully 3.3%.

We conclude that restatement experience corroborates our interpretation of our opacity measure. Firms with higher measures of *OPAQUE* more frequently are required to amend the information they have provided to investors.

4. Regression analysis

In this section, regression analysis is employed to examine the relation between opacity and R^2 as well as crash risk.

4.1. R^2

We begin with results for *IDIOSYN*, which is our measure of the firm-specific information arriving to the securities market. Although several earlier papers have

¹⁰ Discretionary accruals have been widely used elsewhere as a proxy for earnings management. For a review, see Dechow and Schrand (2004, Chapter 4).

¹¹ One potential issue in this analysis is matching a restatement event to the time period in which the accounting irregularities occurred. Existing research demonstrates that on average firms can manipulate accruals for three years before being discovered and having to restate their financial statements (see, e.g., Dechow, Sloan and Sweeney, 1996).

(footnote continued)

Examining the 16 detailed case studies contained in the GAO report confirms this assessment. In 13 out of the 16 cases, (at least) the three prior years' financial statements had to be restated. Because *OPAQUE* is a three-year moving sum of the absolute value of discretionary accruals, it measures firm behavior in an appropriate window prior to any restatement. To the extent that a three-year window is not appropriate for some firms, that would tend only to weaken the association we find between *OPAQUE* and restatements.

explored the relation between opacity and R^2 , our measure of opacity differs from previous work as it is based on characteristics of the individual firm, specifically, on its earnings management. We wish to confirm that our measure of opacity is related to the level of firm-specific information reflected in stock returns.

In Table 5, we find break points that sort our sample into five size quintiles and into three opacity groups (1 = low; 3 = high). With a sample of about 40,800 firm-years, each size quintile contains a bit more than 8,100 firm-years while each opacity group contains about 13,600 firm-years. Panel A reports the number of firms falling into each size-opacity cell. The average value for market capitalization within each size quintile is presented on the left margin. Consistent with the negative correlation between size and opacity reported in Table 3, we observe that for the small-firm quintiles (1 and 2), the number of observations increases moving across rows, indicating that small firms are more likely to be characterized by high opacity. Symmetrically, for the

larger-firm quintiles (4 and 5), the number of firms decreases across each row.

Panel B presents the average value of opacity for each cell in the 5×3 grid. Moving across rows, it is apparent that the sample exhibits considerable variability in opacity even within each size bracket.

Panel C shows how R^2 varies with opacity within various size groups. We know from earlier work (e.g., Roll, 1988) that R^2 varies with size, so it seems crucial to condition on at least this variable. Because size and opacity are also correlated, we wish to confirm that any association between opacity and R^2 is not spurious, i.e., that opacity independently affects R^2 after controlling for firm size. Comparing results across each row, we observe that higher opacity is associated with higher R^2 . The difference in R^2 is small, but small changes in R^2 across differing information regimes are consistent with prior research.¹² The entries on the right margin show significance levels for the test of the hypothesis that the R^2 s in the low- and high-opacity groups are identical. For each size quintile the hypothesis is easily rejected. Thus, opacity does appear to have an independent impact on R^2 .

Table 6 presents regression analysis for *IDIOSYN* that allows us to control for a range of other variables in addition to firm size. The variance of the industry return is mechanically related to R^2 : Higher industry volatility increases explained risk and therefore R^2 , which reduces *IDIOSYN*.¹³ We also include size, the market-to-book ratio, leverage, and contemporaneous return on equity (ROE) as additional controls for firm characteristics.¹⁴ Larger firms operating in a wider cross section of the economy are expected to have higher R^2 s. The market-to-book ratio places firms along a growth-versus-value spectrum and thus could be systematically related to R^2 . Leverage is also expected to affect R^2 through its impact on the sensitivity of firm returns to macroeconomic conditions and because it affects the division of risk between equity and debt holders.

Finally, Jin and Myers include skewness and kurtosis as control variables in their examination of R^2 and opacity. Following their lead, we add these controls in Model 2 of Table 6. Neither of these variations has a meaningful

Table 5

Firm characteristics in groups sorted by size and opacity.

We sort our sample into five size quintiles and into three opacity groups (1 = low; 3 = high). Break points for the two sorts are estimated independently, so the number of observations in each cell differs. Panel A presents the number of observations in each cell. Panel B demonstrates that opacity varies substantially even within size groups. Panel C presents the average R^2 for firm-years in each cell and a test of the hypothesis that R^2 is equal in the high versus low opacity group within each size quintile. There are 40,882 firm-years in the sample period 1991–2005.

Panel A: Observations in each size/opacity group						
Size	Market cap (millions of dollars)	Opacity group			Total	
		1	2	3		
1	39	1,568	2,701	3,902	8,171	
2	123	2,002	2,772	3,406	8,180	
3	331	2,536	2,778	2,865	8,179	
4	994	3,249	2,855	2,076	8,180	
5	14,318	4,267	2,526	1,379	8,172	
Total		13,622	13,632	13,628		
Panel B: Level of opacity for each size/opacity group						
Size		Opacity group				
		1	2	3		
1		0.085	0.184	0.482		
2		0.082	0.181	0.488		
3		0.082	0.178	0.477		
4		0.080	0.176	0.448		
5		0.074	0.175	0.433		
Panel C: R^2 for each size/opacity group						
Size		Opacity group			p-value: (3)–(1)	
		1	2	3		
1		0.147	0.150	0.160	<0.0001	
2		0.177	0.181	0.189	<0.0001	
3		0.224	0.225	0.233	0.0058	
4		0.278	0.284	0.292	0.0011	
5		0.395	0.403	0.417	0.0003	

¹² In the Roll (1988) study, for example, the difference between average R^2 across days in which a firm is or is not mentioned in the financial press is also <0.02.

¹³ Higher market volatility also should increase R^2 , but while this factor varies across years, it is identical for all stocks in any year, whereas industry variability exhibits considerable cross-sectional dispersion. Moreover, the set of year-by-year industry volatilities should span much of the time series variation in market index volatility. Nevertheless, we did estimate Table 6 regressions including market volatility as a right-hand-side variable. Inclusion of this variable had negligible effects on any of our results.

¹⁴ In the accounting literature, studies of the impact of discretionary accruals on stock returns typically include controls for contemporaneous firm performance such as ROE. However, one might prefer to examine stock return characteristics during a given period without allowing for the influence of contemporaneous announcements. We therefore reestimate all of our regressions without ROE as a control variable. As it turns out, the inclusion of ROE has almost no impact on any of the other regression estimates. Panel B of Table 6 contains these and other robustness checks.

Table 6Relation between *IDIOSYN* and opacity.

Panel A contains ordinary least square regressions of *IDIOSYN* as a function of opacity and control variables. *IDIOSYN* measures firm-specific information: $IDIOSYN = \ln[(1-R^2)/R^2]$. There are 40,882 firm-years in the sample period 1991–2005. *t*-statistics appear below regression coefficients. Economic impact is the expected change in firm R^2 resulting from an increase in each right-hand-side variable from the 25th to the 75th percentile of the sample distribution. It is derived by calculating the impact on *IDIOSYN* and inverting to estimate the impact on R^2 . Panel B contains robustness checks for coefficients on *OPAQUE* and *OPAQUE*² for variant specifications of Model 3 from Panel A. See Appendix A for all variable definitions.

Panel A Variable	Model 1	Model 2	Model 3	Economic impact (Model 3)
Intercept	2.870 190.22	2.740 177.52	2.782 160.66	
<i>OPAQUE</i>	−0.163 −8.95	−0.167 −9.34	−0.402 −8.49	0.015
Var(Industry Index)	−125.228 −33.49	−124.669 −33.89	−123.064 −33.35	0.019
Size (lagged)	−0.254 −127.44	−0.247 −122.82	−0.249 −121.89	0.157
<i>M</i> to <i>B</i> (lagged)	0.003 2.51	0.003 2.43	0.003 2.67	−0.001
<i>LEV</i> (lagged)	0.001 0.07	0.006 0.35	0.003 0.15	0.000
ROE	−0.022 −2.43	−0.008 −0.94	−0.006 −0.67	0.000
Skewness		0.030 6.17	0.030 6.06	−0.004
Kurtosis		0.051 37.02	0.051 37.14	−0.018
<i>OPAQUE</i> ²			0.228 5.35	−0.003
Adjusted R^2	0.335	0.357	0.357	
Panel B				
Model variation			Coefficient on <i>OPAQUE</i>	Coefficient on <i>OPAQUE</i> ²
1. Baseline regression (Model 3 Panel A)			−0.402 −8.49	0.228 5.35
2. Excluding ROE from set of explanatory variables			−0.401 −8.48	0.230 5.39
3. Requiring 51 weeks of data per year			−0.391 −8.04	0.223 5.02
4. Measuring opacity using single-year discretionary accruals			−0.462 −5.66	0.554 3.44
5. Using biweekly instead of weekly returns in index model			−0.431 −7.73	0.231 4.58
6. Using percentile rank of <i>OPAQUE</i> instead of its numerical value			−0.099 −7.33	
7. By-year (Fama & MacBeth) regressions, full sample			−0.189 −1.72	0.115 1.53
8. By-year (Fama & MacBeth) regressions, excluding post-2002 data			−0.333 −3.31	0.211 3.16

impact on the relation between opacity and R^2 : The coefficients on the moving sum of absolute discretionary accruals, *OPAQUE*, are −0.163 and −0.167 in Models 1

and 2, respectively. In both cases, the estimate is highly significant, with *t*-statistics of 8.95 and 9.34, respectively.

Consistent with Table 5, firm size is a highly significant predictor of *IDIOSYN*. Roll (1988) also finds that R^2 increases with size, especially in simple index-model regressions (see his Fig. 2). Market-to-book ratio has a statistically significant but economically trivial impact on *IDIOSYN*, and leverage is not at all significant. Not surprisingly, industry variance is highly significant, with high-volatility industries exhibiting lower values of *IDIOSYN* (higher R^2). Both kurtosis and skewness turn out to be statistically significant, with higher values of each implying higher values of *IDIOSYN*. This seems reasonable, because jump events would tend to weaken the link between firm returns and market returns.

In Model 3 of Table 6, we examine potential nonlinearities in the relation between *OPAQUE* and *IDIOSYN*. These could arise from two sources. First, earnings management could have an impact on information flow only when it is relatively aggressive, possibly at the upper end of the observed range of *OPAQUE*. If so, the relation between *OPAQUE* and *IDIOSYN* would be concave, with a steeper reduction in *IDIOSYN* at higher values of *OPAQUE*. Alternatively, we might observe thresholds beyond which opacity is already so high that additional earnings management does not substantially degrade already impaired information flow. If so, we would find that the reduction in *IDIOSYN* as *OPAQUE* increases tails off, resulting in a convex relation between *OPAQUE* and *IDIOSYN*.

The empirical relation appears to be convex. In Model 3 of Table 6, we add the square of *OPAQUE* to the list of explanatory variables. Adding this quadratic term causes the coefficient on *OPAQUE* to more than double to −0.402, but the coefficient on *OPAQUE*² is positive and highly statistically significant (*t*-statistic = 5.35).¹⁵

Fig. 5 plots implied values of R^2 using the parameter estimates from Model 3 as *OPAQUE* moves from the bottom to the top of its sample distribution, holding all other variables at their mean values.¹⁶ R^2 increases by around 13%, from 0.206 to 0.233, from the bottom to the top of the distribution of *OPAQUE*. Given the lack of response in R^2 to explicit news about the firm shown in Roll (1988), a difference of this magnitude could be all that we can expect to uncover with respect to underlying earnings management. Put slightly differently, the quantitative impact of *OPAQUE* on R^2 is as strong as the appearance of explicit firm-specific news in the financial press.

¹⁵ We experiment with higher order terms for other independent variables, specifically, skewness and kurtosis. As size already was a concave (log) transformation of market value, we did not add a nonlinear term for that variable. The inclusion or exclusion of these additional quadratic terms affected neither the coefficients nor the significance levels of *OPAQUE* or *OPAQUE*² in any meaningful manner. Similarly, we experiment with additional squared terms in Table 7 regressions. Again, our inferences on crash risk were unaffected.

¹⁶ We invert Eq. (5) to solve for $R^2 = 1/(\exp(IDIOSYN)+1)$. While *IDIOSYN* is decreasing and convex in *OPAQUE*, R^2 is increasing and concave. The effect of *OPAQUE* on synchronicity tails off as *OPAQUE* increases.

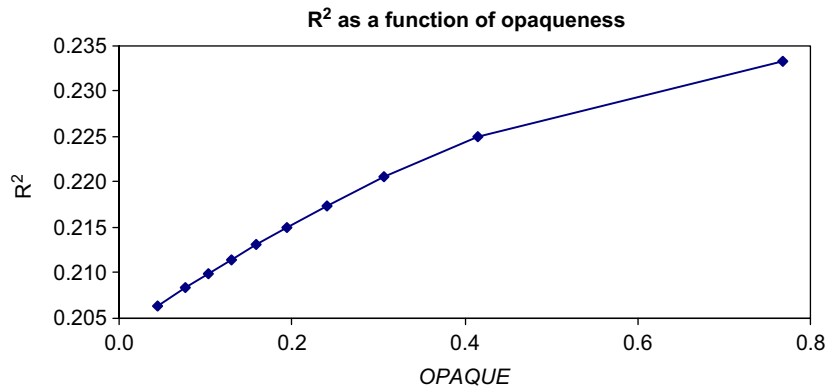


Fig. 5. R^2 as a function of opacity. The horizontal axis value for point on the graph is the average value of *OPAQUE* for decile portfolios formed by ranking on *OPAQUE*. The vertical axis value is the inferred value of R^2 from regression Model 5 of Table 6, with all other right-hand-side variables set equal to their sample average values.

The last column of Table 6 presents estimates of economic impacts of the other right-hand-side variables using coefficient estimates from Model 3, the most inclusive of the specifications. Each entry in that column is the expected impact on R^2 resulting from an increase in the right-hand-side variable from the 25th to the 75th percentile of the sample distribution. [The regression estimates the impact on *IDIOSYN*. We obtain the impact on R^2 by inverting Eq. (5)]. Not surprisingly given the results in Roll (1988), size clearly dominates, but that is in part due to its tremendous sample skewness and correspondingly large interquartile range. Also not surprising, the variance of the industry return has a strong positive impact on R^2 , and kurtosis has a large negative impact. Of most interest to us, the combined economic impact of *OPAQUE* and *OPAQUE*² is $0.015 - 0.003 = 0.012$ or 1.2%. In Roll (1988), the difference between average R^2 across days in which a firm is or is not mentioned in the financial press is only about 2%. In this light, the impact of *OPAQUE* is not trivial.

4.2. Robustness

Our results are robust to reasonable changes in methodology. Panel B of Table 6 shows the coefficient estimates on *OPAQUE* and *OPAQUE*² for a variety of specifications. Line 1 of the panel repeats the estimates from Model 3 of Table 6. Line 2 shows that the coefficients on *OPAQUE* and *OPAQUE*² are barely affected by excluding ROE from the right-hand side. We also experiment with our sample-selection criterion and find, in Line 3, that requiring 51 weeks of returns instead of 26 made virtually no difference in our estimates, presumably because the more stringent criterion results in the loss of few firms from the sample. We try measuring opacity using single-year absolute values of discretionary accruals instead of a three-year moving sum. The regression coefficients (Line 4) are qualitatively similar, but with somewhat lower significance levels, consistent with the interpretation that this is a noisier measure of earnings manipulation. We re-run our index model regressions using two-week instead

of one-week holding periods and find that this makes little difference to our results (Line 5).

We also focus on the model of discretionary accruals. The Jones model could fit accruals behavior better for some industries than others. If so, what appears to be high discretionary accruals and therefore earnings management for some firms might simply reflect higher model error for that industry. Therefore, for each industry we rank order firms by *OPAQUE* and assign each firm its percentile score of *OPAQUE* within its industry. This ordinal ranking automatically adjusts for systematic differences in absolute abnormal accruals across industries. As it turns out, regression coefficients on the ordinal ranking of *OPAQUE* are also negative (Line 6), with still-high levels of statistical significance.¹⁷

A potential problem with the panel regressions in Table 6 is the possibility of within-firm autocorrelation, which would result in biased standard errors. Therefore, we also estimate each regression equation as a cross-sectional regression for each year of the sample independently and compute autocorrelation-corrected Fama and MacBeth (1973) estimates using a method suggested by Pontiff (1996). Specifically, the time-series of the parameter estimates are regressed on a constant, with regression residuals modeled as a fifth-order autoregressive process. The standard error of the intercept term is used as the corrected standard error. As long as the fifth-order autoregressive process captures all of the serial dependence, these standard errors are not biased by serial or cross-sectional correlation.¹⁸ The by-year Fama and MacBeth results (Line 7) are generally consistent with the results in Panel A. However, both the magnitude and the significance levels of the coefficients are considerably reduced. Our analysis (in Table 10) suggests that there is a structural break in the model after the passage of

¹⁷ Given that we now use an ordinal rank, we exclude the square of *OPAQUE* from the right-hand side; Squaring the ordinal rank of opacity would be a meaningless transformation.

¹⁸ An analysis of the significance of the autoregressive parameters indicates that a fifth-order process is well specified. Detailed results for the by-year regressions are available upon request.

Sarbanes-Oxley in 2002 and its likely impact on accounting practice and transparency. Therefore, in Line 8 we repeat the Fama and MacBeth analysis, but ending the sample in 2002. These results are far more in line with the other variants in the table. There are not enough post-Sarbanes-Oxley years to produce a meaningful Fama and MacBeth analysis, but those years contribute considerably to variation in the coefficient estimates in the full sample and therefore reduce the *t*-statistics that appear in Line 7.

4.3. Crash risk

The coefficients on *OPAQUE* in Table 6 suggest that earnings management does impede the flow of firm-specific information to the securities market. We turn next

Table 7

Using opacity to predict crash risk.

In Panel A, logit regressions are used to explain crash risk. The dependent variable, *CRASH* is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations below the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. Panel B contains robustness checks for coefficients on *OPAQUE* and *OPAQUE*² for variant specifications of the Panel A regression. See Appendix A for all variable definitions.

Panel A Variable	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	−2.232	1364.49	<0.0001	
<i>OPAQUE</i>	0.855	25.61	<0.0001	
<i>OPAQUE</i> ²	−0.729	22.11	<0.0001	0.0156
<i>ROE</i> _{<i>t</i>}	−0.213	51.73	<0.0001	−0.0047
<i>SIZE</i> _{<i>t</i>−1}	0.118	278.63	<0.0001	0.0477
<i>M</i> to <i>B</i> _{<i>t</i>−1}	0.007	2.85	0.091	0.0020
<i>LEV</i> _{<i>t</i>−1}	−0.399	39.38	<0.0001	−0.0181
Chi ²	369.53	<0.0001		
Number of observations				
Crash	6,950			
No crash	33,932			
Panel B Model variation	Coefficient on <i>OPAQUE</i> (significance level)	Coefficient on <i>OPAQUE</i> ² (significance level)		
1. Baseline regression (as reported in Panel A)	0.855 <0.0001	−0.729 <0.0001		
2. Excluding <i>ROE</i> from set of explanatory variables	0.882 <0.0001	−0.682 <0.0001		
3. Requiring 51 weeks of data per year	0.942 <0.0001	−0.802 <0.0001		
4. Measuring opacity using single-year discretionary accruals	0.800 0.0067	−0.802 <0.0001		
5. Using biweekly instead of weekly returns in index model	0.804 0.0067	−0.773 0.0047		
6. Using percentile rank of <i>OPAQUE</i> instead of its numerical value	0.083 0.088			
7. By-year (Fama & MacBeth) regressions	0.762 <0.0001	−0.702 <0.0001		
8. By-year (Fama & MacBeth) regressions, excluding post- 2002 data	0.932 <0.0001	−0.851 <0.0001		

to the question of whether such opacity also predicts crash risk.

Table 7 presents logit analysis supporting the link between opacity and crash risk. Each firm-year is assigned a zero if the firm experiences no crash during the fiscal year and a one if there is at least one week during the fiscal year in which the stock price crashes. In light of the strong evidence of nonlinearity in Table 6 regressions, we include both *OPAQUE* and *OPAQUE*² on the right-hand side. The marginal impact of the explanatory variables is presented in the last column of Table 7. Analogous to Table 6, it is the increase in the probability of a crash in any year corresponding to a shift from the 25th to the 75th percentile of the distribution of each right-hand-side variable, while holding all other explanatory variables at their mean value.

Not surprisingly, *ROE* is highly significant with a negative coefficient, implying that firms with good operating performance have lower crash risk during the year. Size is associated with greater crash risk and has the largest marginal impact of any of the explanatory variables. This positive association is largely a function of how we define firm-specific crashes (see Table 8 below). The market-to-book ratio is statistically significant in predicting crash risk in Table 7, but its marginal impact is small. Leverage is associated with lower crash risk. One would think the partial impact of leverage would be to increase crash risk. This contrary finding most likely reflects endogeneity in firms' capital structure choices, specifically, that more stable, less crash-prone firms are more willing or able to establish higher levels of indebtedness.

Turning now to our primary variables of interest, the coefficient on *OPAQUE* is positive and significant at better than a 0.01% level while the coefficient on *OPAQUE*² is negative and also significant at better than a 0.01% level. The concavity of the relation in this table is consistent with the convexity of the relation between *IDIOSYN* and *OPAQUE* found in Table 6. In both cases, the impact of *OPAQUE* tails off at higher values of earnings management. To examine the marginal impact for changes in earnings management we must combine the impacts of *OPAQUE* and *OPAQUE*² because both variables move in lockstep as earnings management increases. The combined marginal impact for the two right-hand-side variables is 0.0156.¹⁹

Given the concavity of the regression relation in Table 7 with respect to *OPAQUE*, the incremental impact of an increase in *OPAQUE* steadily diminishes as its value increases. Fig. 6 presents crash probabilities computed using mean values of *OPAQUE* for each of the ten portfolios formed by ranking on *OPAQUE*, with all other right-hand-

¹⁹ Because crash probability is nonlinear in *OPAQUE* and *OPAQUE*², one cannot simply add the marginal impacts of the two variables (as we did in Table 6, where *IDIOSYN* is a linear function of the explanatory variables). The combined impact on probability is not the sum of the individual impacts that would be obtained by mechanically adding up the impacts of each. We compute the combined impact by setting both *OPAQUE* and *OPAQUE*² to their 25th and 75th percentile values, respectively, and comparing crash probabilities at the two percentile values.

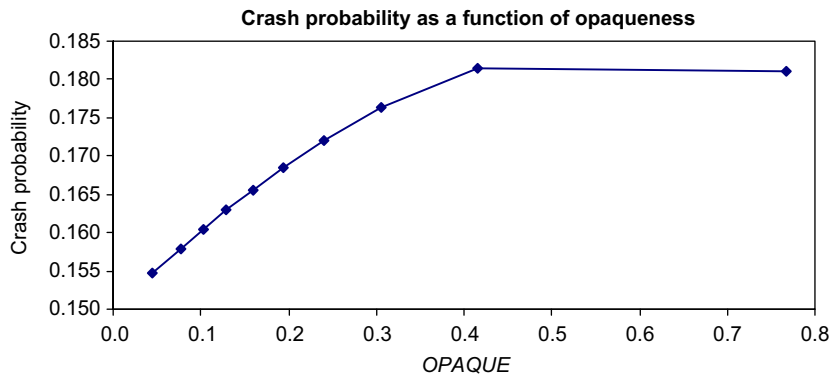


Fig. 6. Annual crash probability as a function of opacity. The horizontal axis value for point on the graph is the average value of *OPAQUE* for decile portfolios formed by ranking on *OPAQUE*. The vertical axis value is the implied probability of a crash during the year for each value of *OPAQUE* based on Model 1 of Table 7, with all other right-hand-side variables set equal to their sample average values.

side variables set equal to their sample means. The sensitivity of crash risk to *OPAQUE* substantially declines at higher values of *OPAQUE*. The change in the implied probability of crashes across the bottom to the top decile is 2.7%.

To summarize, the impact of *OPAQUE* on crash risk is statistically highly significant and consistent with the Jin and Myers (2006) model. The quantitative or economic import of the variable is more difficult to assess. On the one hand, the 2.7 percentage point increase in crash probability that corresponds to the difference between the bottom and top decile of *OPAQUE* is certainly enough to warrant attention in any application focused on tail risk. On the other hand, the unconditional probability of observing a crash during a full year is about 17.1% (Table 2). So our earnings management-based measure of opacity roughly accounts for $2.7/17.1 = 15.8\%$ of the variation in crash risk. On balance, while it appears that the majority of crashes are independent of accounting practice, knowing that a firm is an active earnings manager increases the conditional probability of a crash by an economically as well as statistically significant amount.

We perform similar robustness checks for the determinants of crash risk as for R^2 , and obtain similar results. Panel B of Table 7 shows that the results are robust to the exclusion of ROE on the right-hand side (Line 2), requiring 51 weeks of annual data for each firm (Line 3), to the use of a single-year absolute value of discretionary accruals to measure opacity (Line 4) and to the use of two-week versus one-week holding periods (Line 5). Using the firm's percentile score for *OPAQUE* in its industry does not affect the qualitative conclusion that opacity increases crash risk, although statistical significance is reduced (Line 6). Finally, as in Table 6, Fama and MacBeth coefficients are a bit lower than the panel results, but the relation is stronger when we restrict the sample to the pre-Sarbanes-Oxley years.

For a different look at the magnitude and determinants of crash risk, Table 8, Panel A, presents crash risk for the same size-opacity sort presented in Table 5. No pattern is evident across the rows in the two small-firm quintiles,

but a strong relation characterizes the three larger ones. For the three largest quintiles, the difference in crash probability between the high- and low-opacity groups increases from 3.4 to 4.1, to an astonishing 6.0 percentage points. These are statistically and certainly economically significant differences. Moving down each column the patterns are not as clear, with one exception. In each case, the lowest crash probability is associated with the smallest quintile. This seems counter to casual observation but is consistent with the positive coefficient on size in Table 7, suggesting that large firms appear to be more crash-prone than smaller ones.

This surprising result is an artifact of our definition of a crash. We define a crash as a tail event of sufficient magnitude to fall in the lower 0.1% of the normal distribution. This is roughly a three-sigma event. But large firms have lower standard deviations of returns than smaller ones (the Pearson correlation between size and standard deviation is -0.465 , significant at better than 0.01% level in Table 3). Therefore, the absolute magnitude of a return needed to qualify as a crash is smaller for larger firms. This is evident in Panel B of Table 8, which presents the mean crash return for each size-opacity cell and, in the margins, the mean crash return for each size quintile and opacity group. Moving down the columns, a clear pattern is evident: Raw crash returns are smaller in absolute value for larger firms. The mean crash return is -0.25 for the smallest size quintile and -0.17 for the largest size quintile. Thus, using our definition of a crash, the higher crash propensity of larger firms does not imply necessarily wider swings in returns.

Given this pattern, there is an obvious temptation to define a crash event uniformly across firms, for example, as a three-sigma event based on the average residual standard deviation from the entire sample of firm years. But any such uniform definition would by construction result in high-volatility firms being identified as high-crash firms. In general, to evaluate the jumpiness of any stochastic process, one needs to evaluate movements relative to the standard deviation of that particular process. In our context, crash episodes for each firm must be defined relative to the return volatility of that

Table 8

Crash and jump probabilities for firms sorted on size and opacity.

We sort our sample into five size quintiles and into three opacity groups (1 = low; 3 = high). Break points for the two sorts are estimated independently, so the number of observations in each cell differs. *CRASH* is an indicator variable equal to one if within its fiscal year a firm experiences one or more Firm-Specific Weekly Returns falling 3.09 or more standard deviations below the mean Firm-Specific Weekly Return for its fiscal year and equal to zero otherwise. *JUMP* is an indicator variable equal to one if within its fiscal year a firm experiences one or more Firm-Specific Weekly Returns falling 3.09 or more standard deviations above the mean Firm-Specific Weekly Return for its fiscal year and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005.

Panel A: Crash probability for each size/opacity group						
Size	Opacity group			Difference (3)–(1)	p-value (3)–(1)	
	1	2	3			
1	0.117	0.108	0.108	–0.009	0.362	
2	0.151	0.153	0.158	0.007	0.499	
3	0.174	0.184	0.208	0.034	0.001	
4	0.202	0.221	0.243	0.041	<0.0001	
5	0.161	0.181	0.221	0.060	<0.0001	
Panel B: Mean of crash returns (raw one-week stock returns) for each size/opacity group						
Size	Opacity group					Mean
	1	2	3			
1		–0.22	–0.24	–0.26		–0.25
2		–0.23	–0.25	–0.28		–0.26
3		–0.21	–0.23	–0.29		–0.25
4		–0.19	–0.22	–0.26		–0.22
5		–0.15	–0.18	–0.22		–0.17
Mean		–0.19	–0.22	–0.27		
Panel C: Jump probability for each size/opacity group						
Size	Opacity group			Difference (3)–(1)	p-value (3)–(1)	
	1	2	3			
1	0.337	0.358	0.340	0.003	0.813	
2	0.255	0.257	0.247	–0.008	0.521	
3	0.208	0.208	0.198	–0.010	0.348	
4	0.174	0.168	0.161	–0.013	0.227	
5	0.132	0.143	0.123	–0.009	0.367	

particular firm. By their nature, fat tails can be neither assessed nor identified without reference to a benchmark determined by the volatility of the specific underlying process.

While size and standard deviation are negatively correlated, *OPAQUE* and the volatility of weekly returns are positively correlated (the Pearson correlation between opacity and standard deviation is 0.358, significant at better than a 0.01% level). Therefore, the threshold return that qualifies as a crash is larger in absolute value for firms with greater earnings management. This is also apparent in Panel B of Table 8. Within each size quintile, the absolute value of the mean one-week crash return increases monotonically with opacity. Averaging across size quintiles, the mean crash return for the lowest level of opacity is –0.19 and –0.27 for the highest level of opacity.

Table 9

Using opacity to predict positive jumps.

Logit regressions are used to explain jump probability. The dependent variable, *JUMP*, is an indicator variable equal to one if within its fiscal year a firm experiences one or more Firm-Specific Weekly Returns falling 3.09 or more standard deviations above the mean Firm-Specific Weekly Return for its fiscal year and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

Variable	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	–0.167	9.39	0.002	
<i>OPAQUE</i>	–0.164	1.15	0.283	
<i>OPAQUE</i> ²	0.172	1.59	0.208	–0.0031
<i>ROE_t</i>	–0.030	1.12	0.290	–0.0007
<i>SIZE_{t-1}</i>	–0.214	962.59	<0.0001	–0.0881
<i>M to B_{t-1}</i>	–0.011	7.06	0.008	–0.0033
<i>LEV_{t-1}</i>	0.221	14.86	0.000	0.0100
Chi ²	1,270.48	<0.0001		
Number of observations				
Jump	9,029			
No jump	31,853			

4.4. Positive jump risk

Table 7 indicates that earnings management or the opacity of firms' financial reports predicts crashes. These results immediately raise another question: Does opacity predict fat tails generally or only one-sided exposure to crashes? We define a positive jump symmetrically to a crash, that is, as a 3.09 standard deviation positive value of $\ln(1 + \text{residual weekly return})$, and repeat the logit analysis for jump probability.

The simple sort in Table 8, Panel C, shows no evidence of a relation between jump probability and opacity for any of the size quintiles. Table 9 presents logit analysis of jump probability similar to the results in Table 7 for crash risk. The important result in Table 9 is the striking difference between the power of *OPAQUE* to predict crashes (in Table 7) versus its lack of significance as a predictor of positive jumps. The Chi² statistics for both *OPAQUE* and *OPAQUE*² in Table 9 are exceedingly low. Neither of these variables achieves significance at even a 20% confidence level. Thus, in agreement with the Jin and Myers (2006) model, opacity appears to predict only negative crashes in stock prices.

The control variables in Table 9 are largely mirror images of their values in Table 7, with positive coefficients in Table 7 becoming negative coefficients in Table 9. This is the pattern that one would expect. The interesting exception to this rule is the coefficient on *ROE*, which is also negative (but not even close to statistically significant) in the positive jump regressions. The negative coefficient for contemporaneous *ROE* in both tables is supportive of a broad interpretation of the Jin and Myers (2006) and Kirschenheiter and Melumad (2002) models. In both models, a firm receiving bad news tries to hide the negative news until it is no longer feasible to do so, which ultimately results in a crash. Thus, opaque firms with poor performance are more subject to crashes. This explains

why the coefficient on *ROE* would be negative in Table 7 crash regressions. However, firms experiencing good performance have no incentives to hide such news. Because they would not be expected to stockpile positive information, their stock returns ought to reflect good news more steadily over time. Therefore, the impact of *ROE* on jump probability need not be symmetric to its impact on crash risk. Better performance encourages more revelation and therefore more continuous information release.

4.5. Sarbanes-Oxley and transparency

We use the moving sum of absolute discretionary accruals as a proxy for firm opacity and show that this variable predicts both R^2 and crash risk. But earnings management could wax and wane over time, thus providing a natural experiment to further test the impact of opacity on stock price dynamics. Specifically, consider the potential impact of the Sarbanes-Oxley Act, enacted in 2002. Other papers note that the passage of the act, which substantially increased the penalties for earnings manipulation, materially reduced the incidence of accounting-based earnings management (Cohen, Dey, and Lys, 2008; Graham, Harvey and Rajgopal, 2005). Presumably then, the act would also have reduced the power of discretionary accruals to predict the properties of stock returns.

One could view discretionary accruals derived from residuals in the modified Jones model as the sum of model error plus intentional accruals management. If SOX has been successful in reducing earnings management, then in the post-SOX period residuals from the modified Jones model would be composed more of model error and less of intentional accruals management. It follows that the predictive power of *OPAQUE* for both *IDIOSYN* and crash risk would dissipate in the post-SOX environment. Tables 10 and 11 test this hypothesis.

In Table 10, Models 1 and 2, we introduce two new explanatory variables: a SOX dummy (equal to zero prior to 2002 and equal to one in 2002 and beyond) and an interaction term equal to the product of *OPAQUE* and the SOX dummy. As in Table 6, Models 1 and 2 present specifications with and without kurtosis and skewness as control variables. In both cases, the coefficient on the SOX dummy is negative (−0.309 and −0.334) and highly significant (t -statistics above 25). The negative coefficient on SOX indicates that the baseline level of *IDIOSYN* has shifted down or, equivalently, that typical R^2 s have increased since 2002. It is hard to assign causation here, as several other changes could have affected information flows from firms to investors, most notably Regulation Fair Disclosure.

Of greater interest is the coefficient on the product of *OPAQUE* and the SOX dummy. This coefficient can be interpreted as the change in the coefficient on *OPAQUE* in the post-SOX period. In both specifications, while the coefficient on *OPAQUE* is negative (−0.189 in both models), the coefficient on the interaction term is positive and of slightly greater magnitude (either 0.233 or 0.234). The coefficients on both *OPAQUE* and the interaction term are

Table 10

Relation between *IDIOSYN* and opacity pre- and post-Sarbanes-Oxley Act.

Ordinary least squares regressions of *IDIOSYN* as a function of opacity and control variables. *IDIOSYN* measures firm-specific information. *SOX* is an indicator variable equal to one in 2002 and beyond and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. t -Statistics appear below regression coefficients. See Appendix A for all variable definitions.

Variable	Model 1	Model 2	Model 3
Intercept	2.916 191.14	2.782 179.12	2.833 159.77
<i>SOX</i>	−0.309 −25.06	−0.334 −27.58	−0.399 −23.03
<i>OPAQUE</i>	−0.189 −8.67	−0.189 −8.81	−0.512 −9.20
<i>SOX</i> * <i>OPAQUE</i>	0.233 6.54	0.234 6.70	0.723 7.46
Var(Industry Return)	−135.30 −36.42	−135.76 −37.24	−134.40 −36.76
<i>SIZE</i> (lagged)	−0.243 −120.63	−0.234 −115.78	−0.236 −114.81
<i>M-to-B</i> (lagged)	0.001 1.02	0.001 0.71	0.001 0.89
<i>LEV</i> (lagged)	−0.037 −2.05	−0.035 −1.97	−0.036 −2.04
<i>ROE</i>	−0.030 −3.32	−0.016 −1.77	−0.014 −1.53
Skewness		0.028 5.88	0.028 5.76
Kurtosis		0.054 39.91	0.054 39.99
<i>OPAQUE</i> ²			0.322 6.31
<i>SOX</i> * <i>OPAQUE</i> ²			−0.485 −5.49
Adjusted R^2	0.349	0.374	0.374

highly significant, with t -statistics all > 6. In Model 3, we also add *OPAQUE*² and *OPAQUE*² times the SOX dummy. As we find for *OPAQUE*, the coefficient on *OPAQUE*² times the SOX dummy has the opposite sign and is slightly greater in magnitude than that on *OPAQUE*², suggesting that the impact of earnings management ends after 2002.

Therefore, it seems that the relation between abnormal accruals and *IDIOSYN* largely disappears in the post-SOX period. This is consistent with the hypothesis that, in a period with greater monitoring and scrutiny of accounting practice, earnings management has substantially subsided, and residuals from the modified Jones model no longer proxy for the opacity of firms' financial reports.

Table 11 repeats our logit crash analysis allowing for different impacts of opacity in the pre- and post-SOX periods. The results are highly consistent with Tables 7 and 10. *OPAQUE* and *OPAQUE*² predict crashes with high significance in the pre-SOX period. But, while the coefficients on the post-SOX dummy interaction variables are estimated with far less precision, they are all of opposite sign and comparable magnitude as the pre-SOX

Table 11

Using opacity to predict crash risk, pre- and post-Sarbanes-Oxley Act (SOX).

Logit regressions are used to explain crash risk. The dependent variable, *CRASH*, is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations below the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise. *SOX* is an indicator variable equal to one in 2002 and beyond and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

Variable	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	−2.285	1284.14	<0.0001	
SOX	0.389	41.61	<0.0001	0.0570
OPAQUE	0.972	21.35	<0.0001	
OPAQUE ²	−0.893	20.14	<0.0001	0.0173
SOX * OPAQUE	−0.622	3.37	0.067	
SOX * OPAQUE ²	0.565	3.23	0.072	−0.0102
ROE _t	−0.209	49.08	<0.0001	−0.0046
SIZE _{t−1}	0.105	209.86	<0.0001	0.0421
M to B _{t−1}	0.010	6.11	0.013	0.0030
LEV _{t−1}	−0.357	31.51	<0.0001	−0.0162
Chi ²	477.47	<0.0001		
Number of observations				
Crash	6,950			
No crash	33,932			

Table 12

Using opacity to predict positive jumps, pre- and post-Sarbanes-Oxley Act (SOX).

Logit regressions are used to explain jump probability. The dependent variable, *JUMP*, is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations above the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise. *SOX* is an indicator variable equal to one in 2002 and beyond and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	−0.169	8.87	0.003	
SOX	0.196	11.20	0.001	0.0570
OPAQUE	−0.266	2.13	0.145	
OPAQUE ²	0.239	2.08	0.149	−0.0056
SOX * OPAQUE	0.108	0.12	0.734	
SOX * OPAQUE ²	−0.116	0.16	0.686	0.0023
ROE _t	−0.026	0.87	0.351	−0.0046
SIZE _{t−1}	−0.224	1012.58	<0.0001	0.0421
M to B _{t−1}	−0.009	4.81	0.028	0.0030
LEV _{t−1}	0.248	18.71	<0.0001	−0.0162
Chi ²	1327.45	<0.0001		
Number of observations				
Jump	9,029			
No jump	31,853			

values, implying that opacity does not predict crashes in the post-SOX period. Table 12 examines jump propensities in the pre- and post-SOX periods. Consistent with Table 9, *OPAQUE* is not statistically significant in predicting jumps in either the pre- or the post-SOX periods.

5. Conclusions

We confirm that our firm-specific measure of opacity, the three-year moving sum of the absolute value of discretionary accruals, is a reliable predictor of R^2 . We also examine the more sparsely considered relation between opacity and crash risk. We find that our measure of earnings management reliably predicts such risk. In contrast, positive jumps in stock returns are essentially unrelated to our measure of opacity. Reasonable variation in this measure of opacity changes either R^2 or crash risk by about 15% of their unconditional or baseline levels, but the effect on crash risk is substantially more pronounced for larger firms. While opacity, or at least our proxy for it, is clearly part of a larger story in explaining stock price dynamics, it does appear to be an economically meaningful part of that story, and the statistical relation between it and both R^2 and crash risk is strong. These results are new and consistent with a theoretical model of crash risk based on information management: Firms manage earnings to shelter bad information up to a point, but, once some tipping point is crossed, the information comes out in one fell swoop, which results in a price crash. However, this phenomenon appears to have dissipated in the post-SOX years.

Appendix A. Variable definitions

OPAQUE is the prior three years' moving sum of the absolute value of discretionary accruals. Specifically,

$$OPAQUE = AbsV(DiscAcc_{t-1}) + AbsV(DiscAcc_{t-2}) + AbsV(DiscAcc_{t-3}),$$

where $DiscAcc_t$ is measured using the Modified Jones Model.

Firm-Specific Weekly Return is equal to $\ln(1+residual)$, where the residual is $\varepsilon_{j,t}$ from the expanded index-model regression:

$$r_{j,t} = \alpha_j + \beta_{1j}r_{m,t-1} + \beta_{2j}r_{i,t-1} + \beta_{3j}r_{m,t} + \beta_{4j}r_{i,t} + \beta_{5j}r_{m,t+1} + \beta_{6j}r_{i,t+1} + \varepsilon_{j,t}.$$

IDIOSYN is a measure of the firm-specific information arriving to the securities market based on $(1-R^2)$ from the modified index model regression. Specifically,

$$IDIOSYN = \ln\left(\frac{1-R^2}{R^2}\right).$$

$Std\ Dev\ \ln(1+residual)$ is the standard deviation of the *Firm-Specific Weekly Return* over the fiscal year.

Kurtosis is the kurtosis of the *Firm-Specific Weekly Return* over the fiscal year.

Skewness is the skewness of the *Firm-Specific Weekly Return* over the fiscal year.

CRASH is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations below the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise.

JUMP is an indicator variable equal to one if within its fiscal year a firm experience one or more *Firm-Specific*

Weekly Returns rising 3.09 or more standard deviations above the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise.

SIZE is the natural log of the market value of equity at the beginning of the fiscal year.

M to B is the ratio of the market value of equity to the book value of equity measured at the beginning of the fiscal year.

LEV is the book value of all liabilities scaled by total assets, measured at the beginning of the fiscal year.

ROE is contemporaneous return on equity defined as income before extraordinary items divided by the book value of equity.

VAR(Industry Index) is the variance of the weekly returns of the Fama and French industry index during the firm's fiscal year.

SOX is an indicator variable equal to one in sample years 2002 and beyond and equal to zero otherwise.

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